

PROJECT DOCUMENTATION

M&A Valuation Intelligence Platform

Azure Data Engineering + Analytics Project

Platform Used

Microsoft Azure — Azure Data Factory · ADLS Gen2 · Azure Databricks · Delta Lake ·
Azure SQL Database · Power BI

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1. Project Overview

Investment firms, M&A advisory teams, corporate development departments, and business valuation teams regularly need to understand how companies are valued across industries and enterprise-value ranges. In practice, this valuation data is scattered across multiple files and requires significant manual processing before an analyst can answer even routine questions — what a typical EV/EBITDA multiple looks like for an industry, how multiples have moved over time, or how reliable a given benchmark actually is.

The M&A Valuation Intelligence Platform was built to remove that manual effort. It is a production-style data platform on Microsoft Azure that ingests, validates, stores, transforms, and curates M&A valuation data — ending in a set of Power BI dashboards that let an analyst answer these questions directly, with no spreadsheet wrangling in between.

The platform follows a Medallion Architecture (Bronze → Silver → Gold) built on Azure Data Lake Storage Gen2 and Delta Lake, feeds a relational star schema in Azure SQL Database for fast, familiar querying, and is fronted by a four-page executive Power BI dashboard. Ingestion and transformation — from the public GitHub source through to the curated Gold layer — run automatically on a daily schedule; the Azure SQL serving-layer load and Power BI refresh are currently run on demand.

Objectives

- Automate ingestion of a continuously-updated public valuation dataset, while preserving a full dated history for reproducibility.
- Apply enterprise-grade data quality validation before any data is exposed for analysis.
- Curate business-ready Gold datasets that directly answer the platform's target business questions.
- Expose curated data through a governed relational star schema, including a Slowly Changing Dimension for industry classification.
- Deliver a self-service, interactive Power BI dashboard for executive and analyst consumption.
- Implement the platform securely, with no hard-coded credentials anywhere in the pipeline.

2. Dataset Details

The platform's source is the M&A Multiples Index, a publicly available repository (github.com/nickcals/multiples) that publishes valuation multiple data across industries and enterprise-value brackets. The repository is continuously updated and is treated by the platform as an external, mutable source — its stated source window spans 2018–2026.

The source publishes two structured JSON resources, each with a different grain:

File	Grain	Key Fields
multiples.json	Industry sub-vertical × EV bracket	EV/EBITDA, EV/Revenue, P25 / Median / P75, number of disclosed deals
multiples-by-year.json	Industry sub-vertical × year	EV/EBITDA, EV/Revenue, median deal value, number of deals (time series)

Industry Classification

The source lists 46 industry sub-verticals. Since the source does not provide a higher-level grouping, the platform derives one: each sub-vertical is classified into one of 7 industry groups — Healthcare, Technology, Industrial, Consumer, Energy, Professional Services, and Financial — via a documented keyword-matching rule applied during transformation.

Enterprise-Value Brackets

Five EV brackets are defined by the source: <\$5M, \$5M–\$25M, \$25M–\$100M, \$100M–\$500M, and >\$500M. Because these do not sort correctly as text, the platform assigns each an explicit numeric `bracket_order` (1–5) so Power BI displays them in the correct business order rather than alphabetically.

Reliability & Outlier Rules (per source methodology)

- A published cell must be backed by at least 10 disclosed deals to be considered reliable.
- EV/EBITDA values are expected to fall within 0.5 – 60.0.
- EV/Revenue values are expected to fall within 0.05 – 25.0.

These rules are enforced directly in the platform's data quality framework (Section 8) rather than left as documentation only.

3. Architecture

The platform is organized into five stages, moving data from the public source through to the executive dashboard. Each stage is isolated by an Azure service boundary, so any stage can be re-run, monitored, or extended independently.

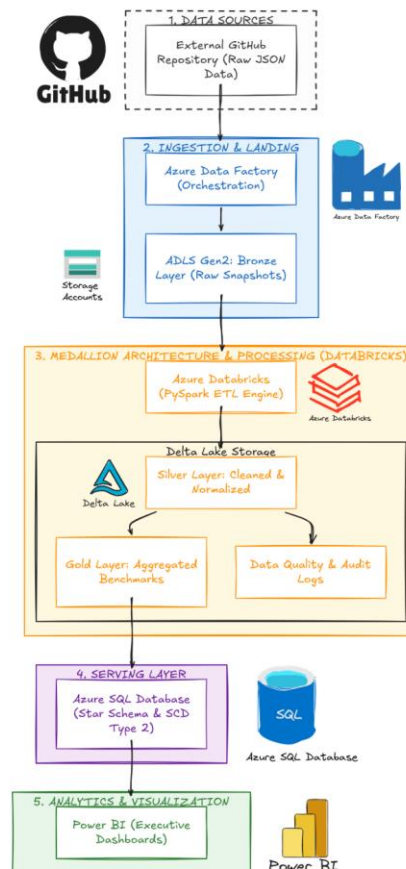


Figure: End-to-end platform architecture on Microsoft Azure

- Data Sources — the external GitHub repository publishing raw JSON valuation data.
- Ingestion & Landing — Azure Data Factory orchestrates retrieval and lands dated snapshots in the ADLS Gen2 Bronze layer.
- Medallion Architecture & Processing — Azure Databricks (PySpark) transforms Bronze into Silver (cleaned) and Gold (curated) Delta tables, with a dedicated data-quality and audit stage.
- Serving Layer — Gold data is loaded into an Azure SQL star schema, including Slowly Changing Dimension logic.

- Analytics & Visualization — Power BI connects to the star schema and serves a four-page executive dashboard.

4. Infrastructure & Environment Setup

All resources were provisioned under a single resource group, keeping the environment easy to manage, monitor, and tear down as a unit.

Resource	Name
Resource Group	rg-ma-valuation-intelligence-platform
Storage Account (ADLS Gen2)	stmavaluationplatform — containers raw, silver, gold, adf-staging
Key Vault	kv-ma-valuation-platform (RBAC permission model)
Azure Data Factory	adf-ma-valuation-intelligence-platform
Azure SQL Server / Database	sql-ma-valuation-platform / ma-valuation-db
Databricks Workspace	dbw-ma-valuation-platform (Unity Catalog enabled)
Databricks Cluster	cluster-ma-valuation (single node, 20-min auto-terminate)
Databricks Job	job-ma-valuation-pipeline

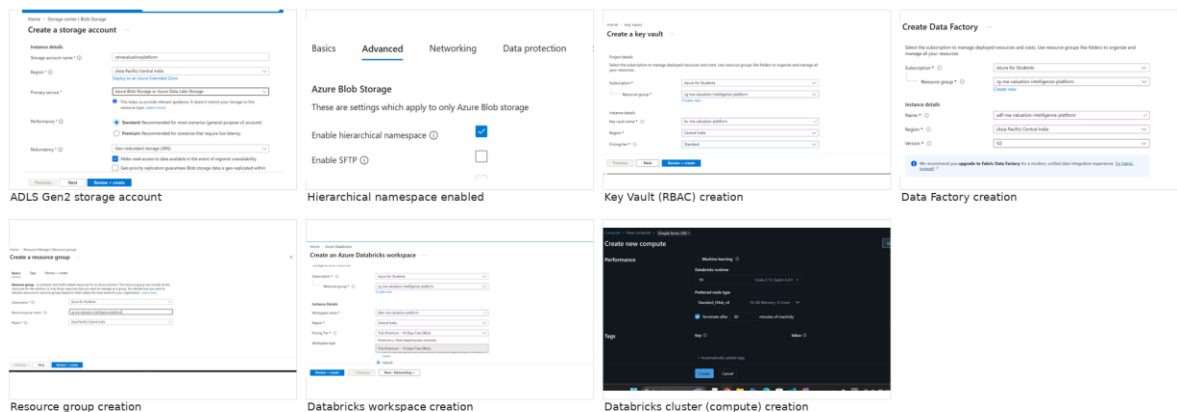


Figure: Core Azure services provisioned — storage account, hierarchical namespace, Key Vault, Data Factory, resource group, and Databricks workspace/cluster

5. Security Implementation

Security was treated as a first-class requirement rather than an afterthought. The platform authenticates using Managed Identity and Microsoft Entra ID throughout — no passwords or storage account keys are stored in any pipeline definition or notebook.

- Azure Key Vault (RBAC permission model) holds the platform's one unavoidable secret — a Databricks personal access token required only because one specific ADF connector type does not support Managed Identity auth.
- A dedicated service principal (sp-databricks-sql-writer) was registered for the one connector that required token-based auth, rather than reusing a broader identity.
- Azure SQL Database uses Microsoft Entra-only authentication — there are no SQL logins on the server.
- Access from Azure Data Factory to SQL, storage, and Databricks is granted via managed identity role assignments, verified directly with a SQL query confirming db_datareader / db_datawriter membership.

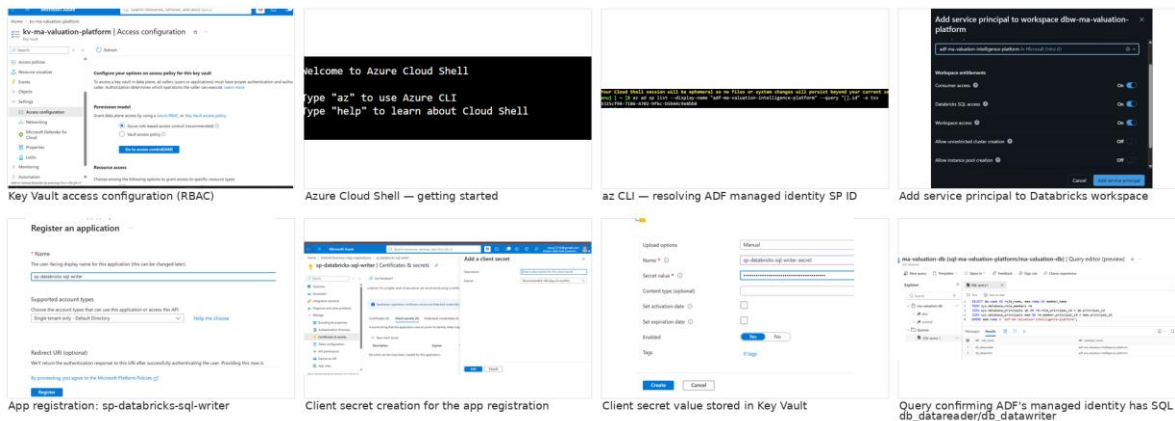


Figure: Key Vault RBAC configuration, service principal registration, client secret storage, and verification of ADF's managed identity permissions in Azure SQL

6. Data Ingestion Layer (Azure Data Factory)

Azure Data Factory acts as the platform's orchestration layer. The pipeline PL_MA_Multiples_Ingestion is fully metadata-driven and parameterized — no source URLs or paths are hard-coded into the pipeline itself.

Pipeline Flow

- SCR_AuditStart logs the start of the run to control.pipeline_audit.
- LKP_GetActiveSources reads the active sources, their URLs, target paths, and load type (FULL or INCREMENTAL) from control.etl_metadata.

- FE_ProcessSources loops over each active source. A Get Metadata check plus a conditional Copy activity ensures a FULL source always re-copies, while an INCREMENTAL source only copies if today's dated file does not already exist.
- Retrieved files land at raw/ma_multiples/ingestion_date=YYYY-MM-DD/, preserving a full dated history rather than overwriting the previous load — giving the platform historical reproducibility.
- On success, a Databricks Job activity automatically triggers the full transformation pipeline, so ingestion and transformation run as one continuous, unattended process.
- SCR_AuditSuccess / SCR_AuditFailure close out the run in the audit log, capturing status, timing, and record counts.

A daily schedule trigger (TR_Schedule_Daily) automates the run, the Copy activity retries transient failures against the external source (3 retries, 30-second interval), and a run_date pipeline parameter supports backfilling a specific historical date.

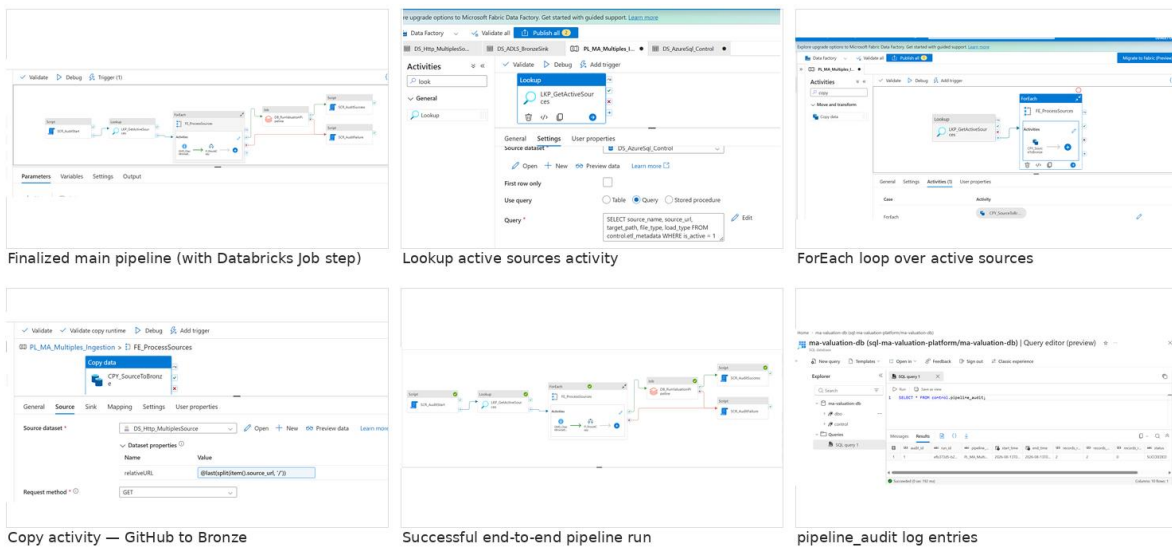


Figure: PL_MA_Multiples_Ingestion pipeline — lookup, ForEach loop, Copy activity, a full successful run, and the resulting pipeline_audit log entry

Triggers, Datasets & Linked Services

Six linked services connect the pipeline to Key Vault, ADLS Gen2, Azure SQL, the public GitHub source, and the two Databricks connector types. Datasets are defined for each source and sink, and the daily schedule trigger is published and actively running.

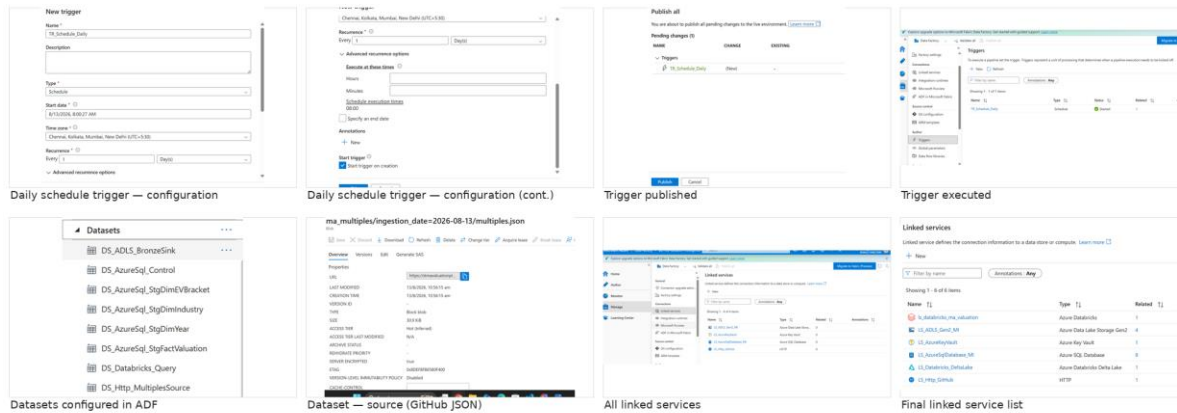


Figure: Daily schedule trigger configuration, published and executed trigger, configured datasets, and the final linked service list

7. Data Transformation (Azure Databricks / Delta Lake)

Azure Databricks is responsible for schema enforcement, JSON parsing and flattening, cleansing, validation, deduplication, standardization, and business transformation — all implemented in PySpark against Delta Lake tables. Five notebooks are chained with linear dependencies into a single Databricks Job, job-ma-valuation-pipeline.

- **01_ingestion_validation** — confirms the Bronze files landed correctly, parses the JSON, and captures row/column counts and source metadata (methodology text, minimum deal-count threshold, schema version).
- **02_bronze_to_silver** — recursively flattens the nested JSON (sub-vertical, bracket, and year are dynamic structure keys, so a Python dict-walk is used rather than Spark's explode) into silver_valuation_multiple_staging, silver_ev_bracket, and silver_industry.
- **03_data_quality** — applies the platform's six validation rules to the staged Silver data, producing a row-level validated table plus an append-only gold_data_quality report table.
- **04_delta_merge** — performs an idempotent MERGE from the validated staging table into the canonical silver_valuation_multiple table, keyed on sub_vertical + metric_type + ev_bracket + source_year, so re-running the pipeline never duplicates data.
- **05_silver_to_gold** — builds four curated Gold Delta tables designed specifically for business consumption: gold_industry_valuation, gold_valuation_trend, gold_industry_ranking, and gold_ev_bracket_analysis.

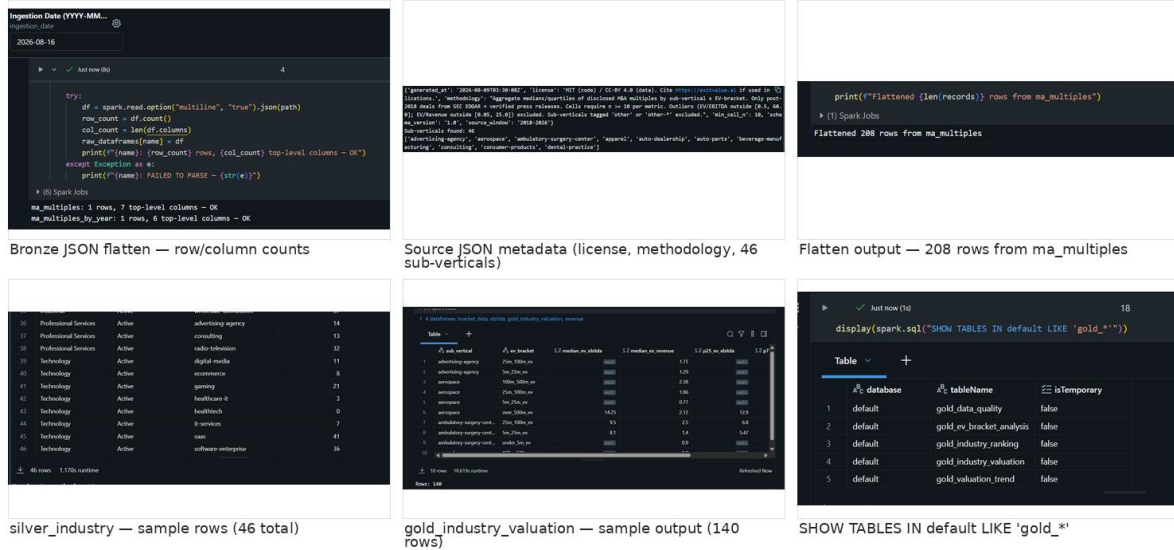


Figure: Bronze JSON flatten output, source metadata capture, flattened record counts, a sample of silver_industry, sample Gold output, and the final set of Gold tables

8. Data Quality Framework

Data quality is a mandatory, automated stage of the pipeline rather than an optional cleanup step. Six rules are applied to every run:

Rule	Description
Null validation	sub_vertical and ev_bracket must not be null
Multiple validation	EV/EBITDA and EV/Revenue must be greater than 0
Percentile consistency	$P25 \leq \text{Median} (P50) \leq P75$
Deal-count validation	deal_count ≥ 10 , per the source's own disclosure threshold
Duplicate validation	Uniqueness enforced on sub_vertical + ev_bracket + metric_type + year
Range / outlier validation	EV/EBITDA within 0.5–60.0; EV/Revenue within 0.05–25.0

Results are written to gold_data_quality — an append-only Delta table tracking run ID, total/valid/invalid/duplicate/null record counts, an overall quality score, and pipeline status for every execution, giving the platform a full audit trail of data health over time.

9. Serving Layer — Azure SQL Star Schema

Rather than exposing the entire Bronze or Silver dataset, the platform loads only business-ready Gold data into a dimensional star schema in Azure SQL — giving analysts a fast, familiar relational model to query directly or connect Power BI to.

- **DimIndustry** — implemented as a full Slowly Changing Dimension (SCD Type 2), tracking `effective_from`, `effective_to`, and `is_current`, so historical industry classifications are preserved even as they evolve.
- **DimEVBracket** and **DimYear** — conformed dimensions sourced from the Silver bracket reference table and the distinct years present in Gold.
- **DimMetric** — a small dimension distinguishing EV/EBITDA from EV/Revenue, used to drive dynamic metric selection in Power BI.
- **FactValuation** — the central fact table, unpivoted from the wide Gold tables into narrow rows keyed by metric, storing P25, median, P75, and deal count against all four dimensions.

The load is implemented as three dedicated ADF pipelines, kept separate from the ingestion pipeline: `PL_Load_Dim_EVBracket_Year`, `PL_Load_Dim_Industry` (SCD Type 2 UPDATE-then-INSERT logic), and `PL_Load_FactValuation` (a multi-branch unpivot query with surrogate-key resolution against all four dimensions). Each stages Delta data through an intermediate ADLS container before loading Azure SQL.

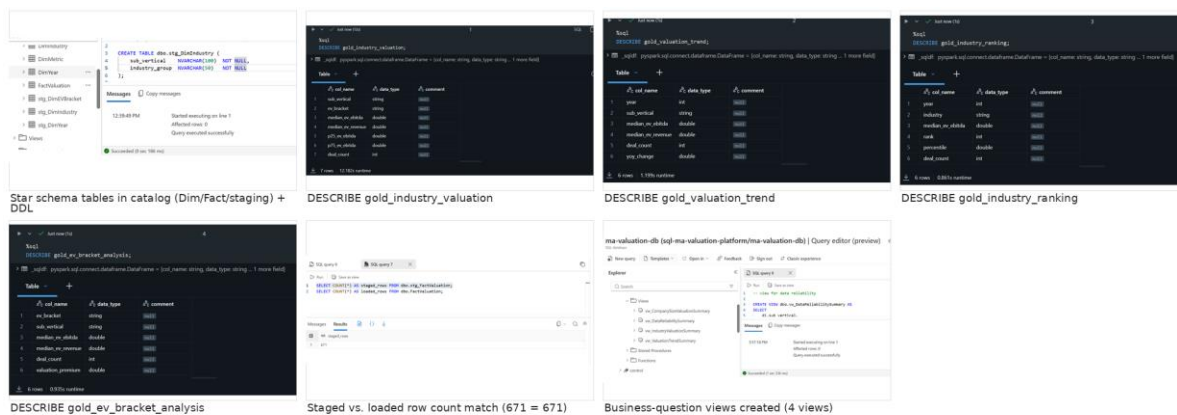
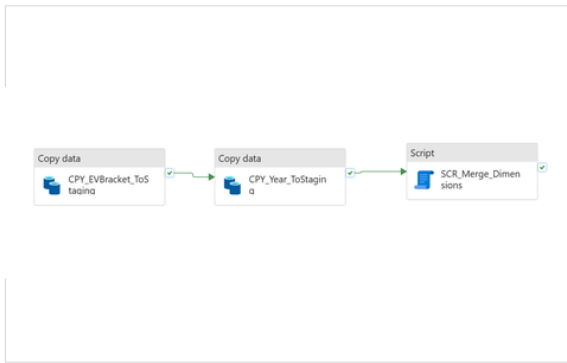
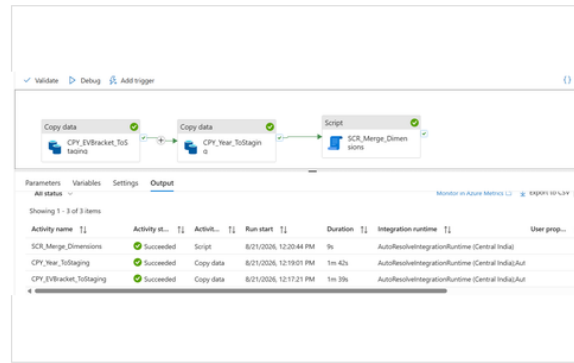


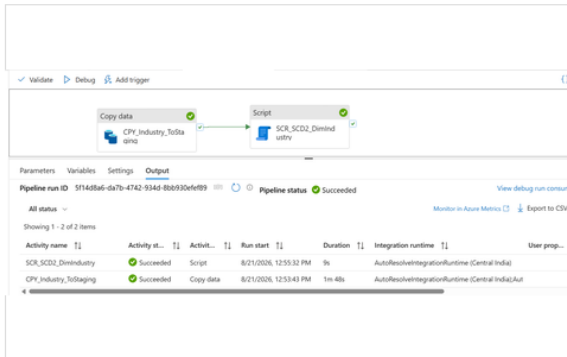
Figure: Star schema tables in the catalog, schema verification for each Gold table, staged-vs-loaded row count reconciliation, and the four business-question views



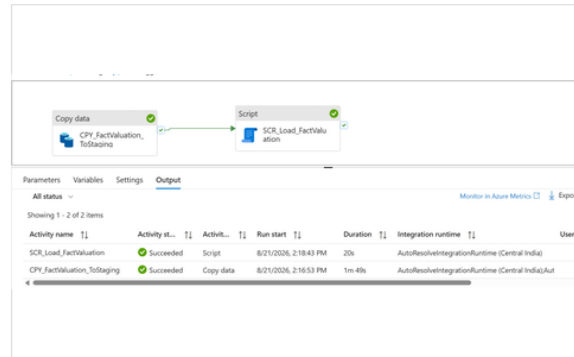
Pipeline: load DimEVBracket + DimYear



Successful run — DimEVBracket + DimYear



Successful run — DimIndustry (SCD Type 2)



Successful run — FactValuation

Figure: The three serving-layer ADF pipelines and their successful runs — DimEVBracket/DimYear, DimIndustry (SCD Type 2), and FactValuation

10. Business Questions Answered

The platform was built to directly answer the business questions defined for the project, organized into four categories. Each category is backed by a dedicated SQL view over the star schema, plus individual queries with evidenced results.

Category	Representative Questions
Industry	Which industries have the highest / lowest median EV/EBITDA? Which have the largest valuation spread?
Company Size	How does EV bracket affect valuation? Do larger companies consistently receive higher multiples?
Time	How have valuation multiples changed since 2018? Which industries saw the largest YoY increase or decline?
Data Reliability	Which benchmarks are backed by the most disclosed deals? Which combinations have low sample sizes?

Each category is implemented as its own SQL view (vw_IndustryValuationSummary, vw_CompanySizeValuationSummary, vw_ValuationTrendSummary, vw_DataReliabilitySummary), with individual runnable queries and captured results demonstrating that every question is genuinely answerable from the platform — not just theoretically supported by the schema.

As of Now — What the Live Dashboard Shows

Based on the current Gold/star-schema load (latest data year 2025, 46 industries, roughly 5,000 disclosed deals, average EV/EBITDA of 12.19x and average EV/Revenue of 2.04x across all industries), the dashboard already surfaces direct answers:

- **Industry** — as of now, healthtech is the highest-valuation industry (Executive Overview KPI). Full lowest-multiple and spread rankings are available live on the Industry Benchmark and Deal Market Intelligence pages.
- **Company Size** — as of now, the Industry Benchmark page's percentile-range (P25/Median/P75) chart lets you compare EV/EBITDA across EV brackets for any industry, showing directly whether larger deal sizes command higher multiples.
- **Time** — as of now, the Valuation Trend page plots the median multiple and year-over-year % change through the latest available year, 2025.
- **Data Reliability** — as of now, the Deal Market Intelligence page's Top-10 ranking (minimum 20-deal filter) and deal-volume chart identify which industry/EV-bracket benchmarks are backed by the strongest sample sizes out of the ~5,000 disclosed deals in the dataset.

11. Power BI Dashboard

Power BI Desktop connects to ma-valuation-db in Import mode, loading the four dimensions and FactValuation. A dedicated Entra ID user (rather than the tenant's owner account) is used for the connection, since Power BI's sign-in flow does not accept the personal Microsoft account used elsewhere in the project.

An 11-measure table drives the dashboard, including fixed KPIs (Total Industries, Total Deal Count, Average EV/EBITDA, Average EV/Revenue) and slicer-driven dynamic measures that respond to a DimMetric slicer — letting the user toggle the entire dashboard between EV/EBITDA and EV/Revenue views.

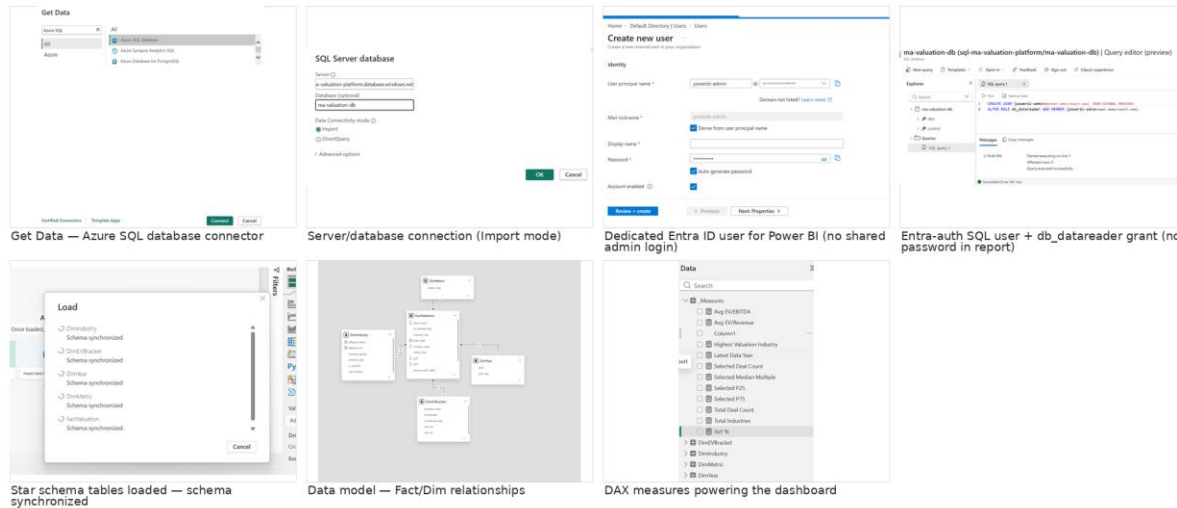


Figure: Connecting Power BI to the Azure SQL serving layer — the Import-mode connection, a dedicated Entra ID user, the loaded star schema, its relationships, and the DAX measures table

Dashboard Pages

- **Page 1 — Executive Overview:** six KPI cards alongside industry valuation ranking, EV bracket distribution, EV/EBITDA vs. EV/Revenue, and deal count by industry.
- **Page 2 — Industry Benchmark:** filterable by industry, EV bracket, year, and metric, with a benchmark table and a percentile-range (P25 / Median / P75) chart per industry.
- **Page 3 — Valuation Trend:** median valuation multiple over time alongside year-over-year percentage change.
- **Page 4 — Deal Market Intelligence:** deal volume by industry, an industry × EV-bracket valuation heatmap, and a Top-10 industries ranking with a minimum deal-count filter layered on top of a Top-N filter.

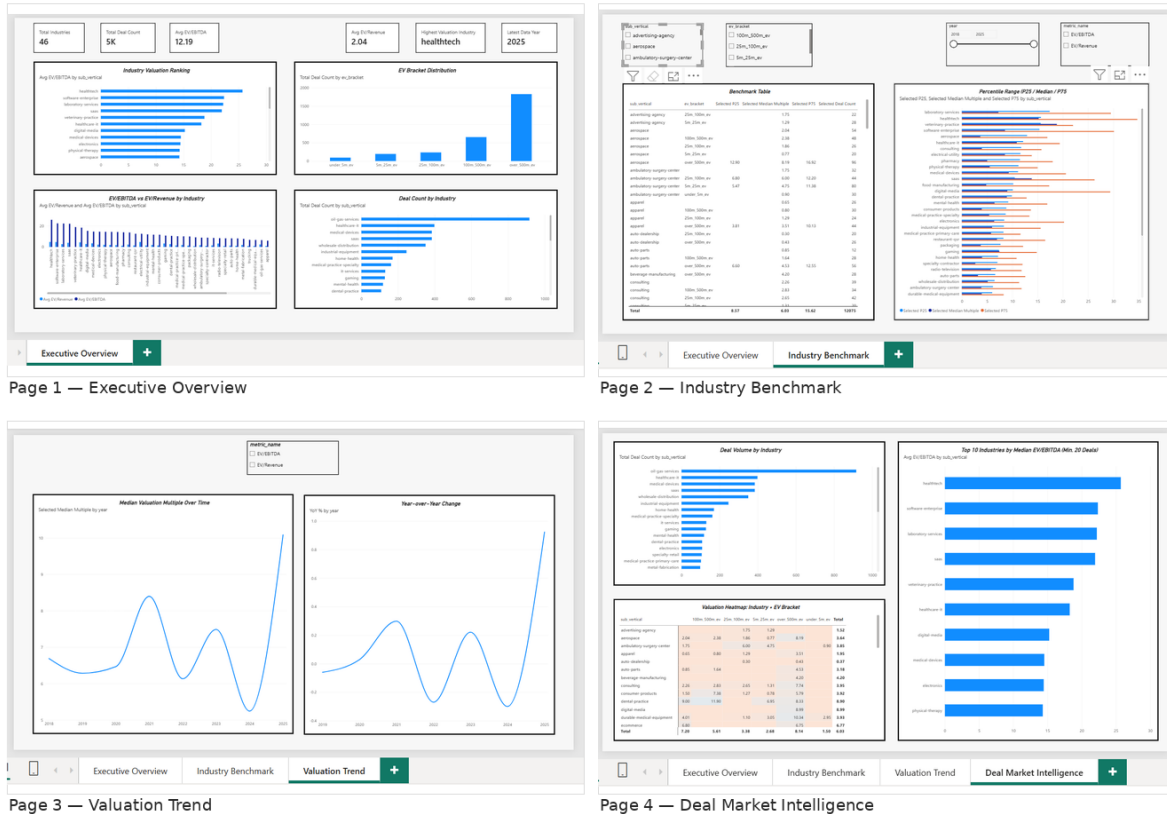


Figure: All four completed Power BI dashboard pages — Executive Overview, Industry Benchmark, Valuation Trend, and Deal Market Intelligence

12. Challenges Faced & Solutions

Several genuine platform-level constraints were encountered during the build — not stylistic choices, but real limitations of the tools involved. Each is documented here with its root cause and the fix that was applied.

- **Nested If Condition activities unsupported** — ADF does not allow an If Condition activity inside the branch of another. Fixed by flattening the logic into a single condition using a combined boolean expression.
- **NULL = NULL breaking duplicate detection and MERGE joins** — SQL's three-valued logic meant nullable key columns (by design, since each row only ever has an EV bracket or a year, not both) never matched. Fixed with a window-function approach for deduplication and an explicit IS NULL fallback in the MERGE ON clause.
- **Silver overwrite silently destroying MERGE history** — an early version of the Bronze-to-Silver notebook overwrote the canonical Silver table directly, wiping

the history the merge step depended on. Fixed by introducing a staging table that only the dedicated merge step writes past.

- **Databricks JDBC and ODBC both failing for the SQL load** — Spark Connect's execution model broke JDBC's low-level driver access, and ODBC driver installation was repeatedly blocked by workspace init-script restrictions. Fixed by abandoning a direct Databricks-to-SQL load entirely and moving the load into ADF Copy and Script activities instead.
- **ADF's Delta Lake connector cannot write directly to Azure SQL** — a hard connector-level requirement. Fixed by staging every load through an intermediate ADLS container before the final SQL write.
- **Two separate Databricks linked-service types required** — the linked service used to trigger the Databricks Job could not also read Delta tables directly. Fixed by creating a second, PAT-based linked service specifically for Delta Lake access.
- **Power BI Desktop rejecting the project's Azure account** — Power BI's SQL connector routes personal Microsoft accounts through a stricter, organizations-only sign-in endpoint. Fixed by creating a dedicated native Microsoft Entra user for Power BI to sign in with.

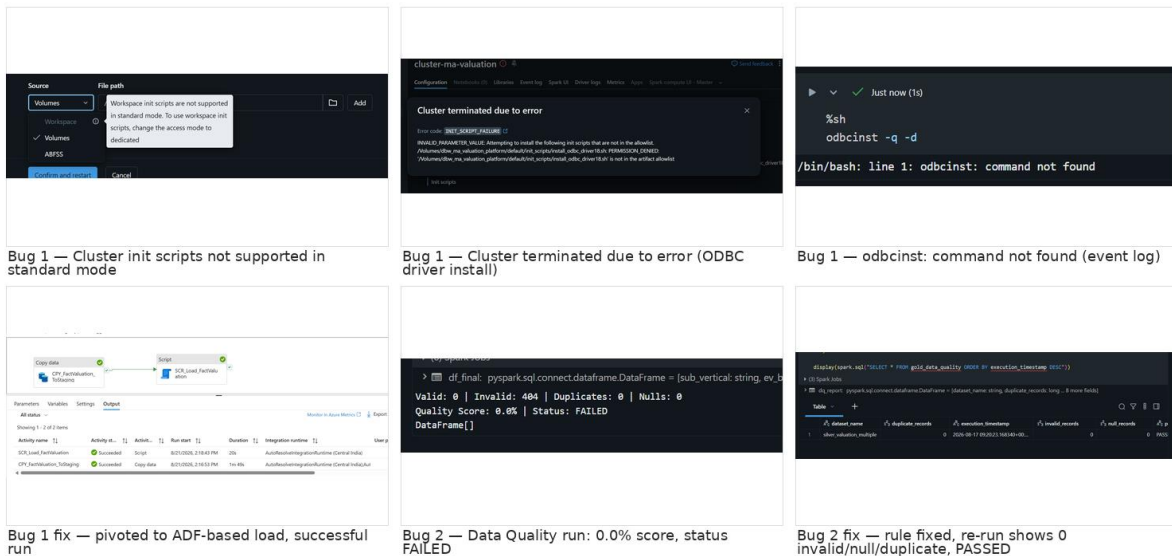


Figure: Evidence of two representative issues and their resolutions — the ODBC driver installation failure that led to the ADF-based load, and the data quality rule fix that brought a failed run to a 100% pass rate

13. Conclusion

The M&A Valuation Intelligence Platform is a complete, product-standard data engineering build on Microsoft Azure — from a raw, continuously-changing public dataset

through to a governed, curated, and secure analytics layer. It applies real Medallion Architecture principles, enforces data quality as a first-class pipeline stage, implements dimensional modeling with a genuine Slowly Changing Dimension, and delivers a dashboard that directly answers the business questions the project set out to solve. Ingestion and transformation run unattended on a daily schedule, with the Azure SQL star schema and Power BI dashboard currently refreshed on demand.